

HEALTHCONNECT CLINIC

Week 6 — Cross-Track Collaboration Document

Track: Data Analytics | Collaboration: Fatimah Odumyiwa - Data Science

1. Purpose of the Collaboration

The collaboration was used to validate the Week 5 reminder/no-show finding and compare analytical processing choices between Data Analytics and Data Science.

2. Shared Starting Point

- Both tracks started from the same raw HealthConnect dataset.
- Both tracks analysed appointment outcomes and reminder information.
- The Data Analytics workflow retained Cancelled appointments in the primary dataset.

3. Methodological Difference Identified

The Data Science collaborator reported 1,285 missing Reminder Channel values, while the Data Analytics workflow had 1,633 missing values at the corresponding stage. The collaborator explained that they excluded 263 Cancelled appointments before analysis, treating cancellations separately from the Attended-versus-No-show comparison. This processing choice explains a substantial part of the difference in sample size and can also change the resulting no-show rate and reminder effect.

The Data Analytics approach did not overwrite the original analysis to force agreement. Instead, Week 6 introduced a separate validation KPI that excludes Cancelled appointments only from the denominator when comparing Attended and No-show outcomes.

4. Data Science Validation Insight

The Data Science collaborator checked the reminder pattern and found a smaller effect: approximately a 5-percentage-point difference in no-show rate, rather than the stronger difference suggested by the initial Data Analytics interpretation. The collaborator explicitly linked the difference to the treatment of Cancelled appointments.

This feedback changed the interpretation of the Week 5 finding. The Week 6 conclusion is now that reminder status is associated with appointment outcomes, but the magnitude of the relationship is sensitive to outcome definition and should not be interpreted as causal evidence.

5. Analytical Contribution to Data Science

- Previous No Shows: identified as a strong candidate feature because current no-show rates rise with previous no-show history.
- Reminder Channel: identified as a candidate feature, but channel effects should be validated while accounting for appointment type and outcome definition.
- Appointment Type: Follow-up shows an elevated 51.23% no-show rate and is a relevant segmentation variable.

- Distance and Waiting Time: Week 5 evidence was weak, so these variables should be empirically tested rather than assumed to be major drivers.

6. Change / Result from Collaboration

The collaboration produced a meaningful analytical refinement: the reminder finding was not simply repeated as a strong effect. Instead, the team identified the source of the discrepancy, added a comparable 51.15% Attended-versus-No-show KPI, and documented the need for consistent outcome definitions in future modelling and evaluation.

7. Recommended Cross-Track Next Steps

- Data Science should test previous no-show history, reminder information, and appointment type as candidate modelling variables.
- Both tracks should use a clearly documented rule for handling Cancelled appointments when evaluating no-show prediction.
- Reminder effects should be validated with consistent denominators and, where possible, stronger evidence than simple observational comparison.
- Week 7 results should be compared across tracks before final operational recommendations are made.

8. Collaboration Limitation

The available collaboration evidence supports methodological comparison and analytical contribution. No predictive model performance improvement is claimed here because a model result was not provided. Any future modelling impact should be documented only after the Data Science track reports the corresponding test or model result.

9. Final Collaboration Statement

The Data Analytics contribution to Data Science is the provision of validated, decision-relevant analytical signals and clarification of their reliability. The collaboration specifically improved interpretation of the reminder finding by exposing the impact of Cancelled-appointment treatment and by identifying candidate features for deeper modelling.